



BATHSHEVA LONDON



Faro

Prototype build manual

INTRODUCTION

Before you begin

From the box of printed parts to a finished, glowing prototype ready to photograph.

Faro is built from seventeen printed pieces, finished by hand in four colours and glued together in a set order. Nothing here is difficult, but each stage needs patience: thin coats of paint, time to dry, and a dry fit before any glue.

TIMING

Weekend one

Inspect and dry-fit, prepare the surfaces, prime, and paint the colours.

The week between

Let the paint harden. It feels dry within hours but needs days to cure.

Weekend two

Clear coat, assemble, light it and photograph.

SAFETY

- Spray outdoors or somewhere very well ventilated, and wear the respirator whenever you spray.
- Wear nitrile gloves when sanding resin and when mixing or applying epoxy.
- Keep sprays and epoxy away from children and pets, and let fumes clear before bringing parts indoors.

HOW TO USE THIS MANUAL

Work through the seven steps in order. Each part's colour and treatment is listed on the next page; keep it open while you paint. If something goes wrong, turn to Troubleshooting at the back: almost every mistake can be sanded out and repainted.



The lantern, finished.

INVENTORY

Parts and supplies

PART	QTY	FINISH
 base	1	Sponged wood effect
 base_plate	1	Black acrylic, or leave primed
 nameplate	1	Gold, black round the letters
 band_cream	1	Cream gloss, clear coat
 band_red	1	Red gloss, clear coat
 tower	1	Cream gloss, clear coat
 knob	1	Gold
 gallery	1	Gold, light coats only
 lantern_frame	1	Gold, light coats only
 cap	1	Red gloss, clear coat
 finial	1	Gold
 lantern_glass	1	Leave unpainted
 window_diffuser_1 to 5	5	Leave unpainted

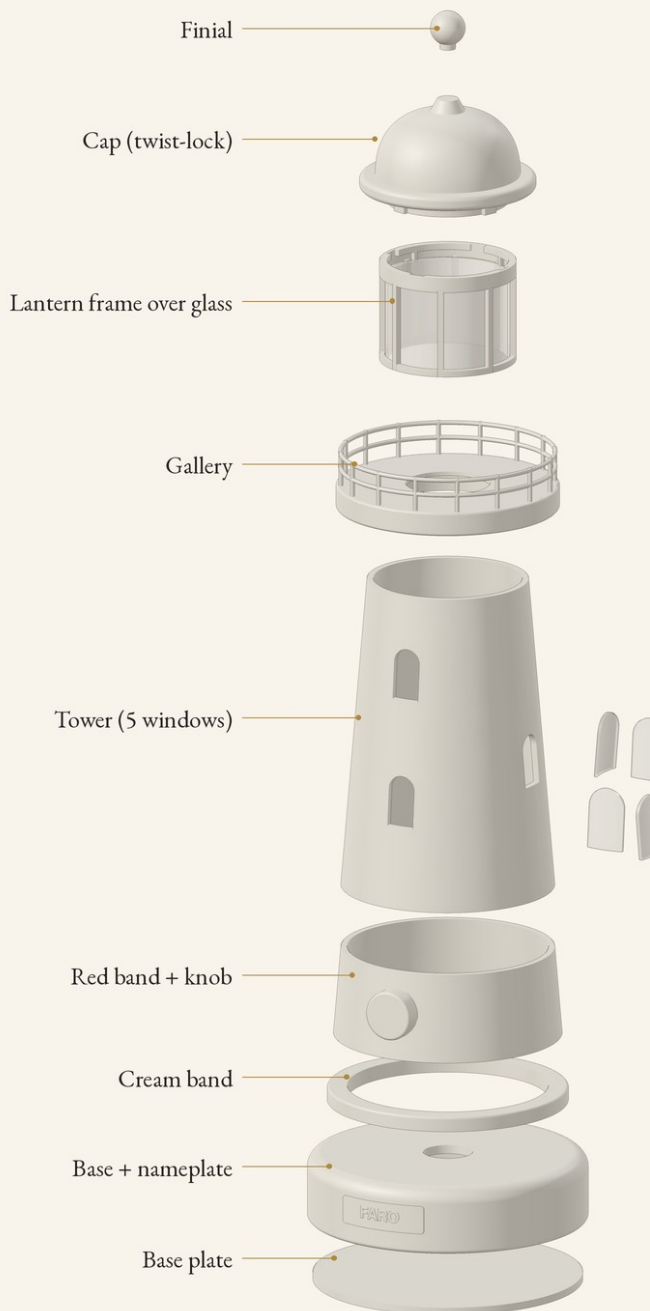
SUPPLIES

- Grey filler primer
- Red gloss spray
- Cream gloss spray (Ivory Bisque)
- Metallic gold spray
- Clear lacquer
- Black and burnt umber acrylic, small brush and sponge
- Wet-and-dry paper, P240 to P1200
- Tamiya masking tape, 10 mm
- Araldite Rapid epoxy, cocktail sticks
- Black self-adhesive felt
- Warm white copper fairy lights, 3 AA batteries
- Rechargeable warm white LED puck, under 45 mm
- A few coins for weight
- Respirator, nitrile gloves, craft knife, blu-tack

STEP 01

Inspect and dry-fit

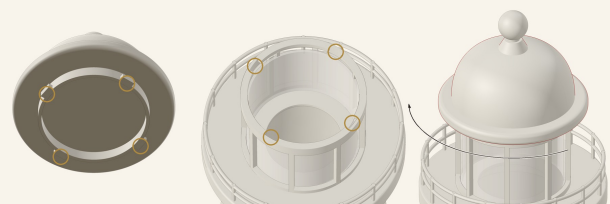
Check every part and fit them together without glue. Paint adds thickness, so tight fits only get tighter.



- 01 Tick off every part against the inventory. Look for cracks, warping or broken railing posts.
- 02 Stack without glue: base, cream band, red band, tower, gallery. Each joint should sit flat and centred.
- 03 Stand the lantern glass on the gallery and lower the frame over it. It should sit without forcing.
- 04 Twist the cap on: lugs down through the slots, then clockwise to the stop.
- 05 Hold each numbered diffuser behind its window. They are not interchangeable.
- 06 Press the base plate into the rebate under the base.
- 07 Stand it where it would live and judge size and proportions.



Dry-stacked: knob and nameplate aligned on the front.



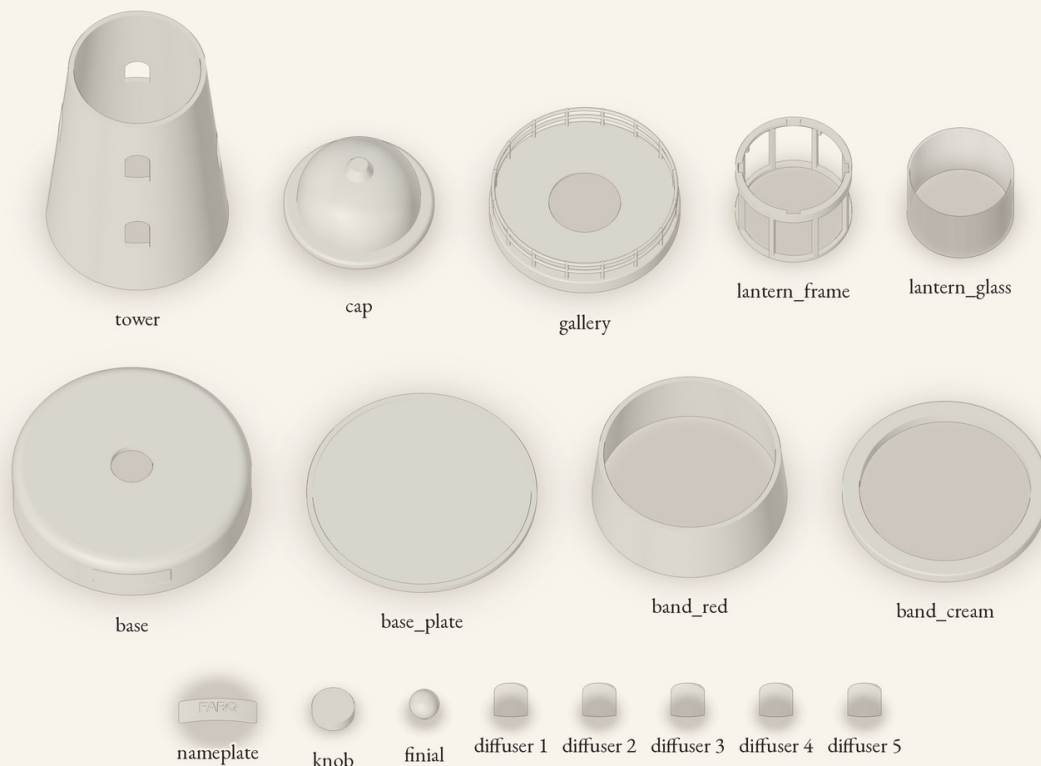
The cap drops in and turns clockwise to lock.

STEP 02

Prepare the surfaces

Clean, smooth surfaces are what make the paint look expensive.

- 01 Wash all parts in warm soapy water, rinse, and leave to dry fully.
- 02 Trim any support nubs with a craft knife, cutting away from yourself.
- 03 Sand visible surfaces with P240 to flatten marks, then P400 to smooth.
- 04 Wipe the dust off with a barely damp cloth and let dry.



Every printed part, as it arrives in plain white resin.

HANDLE WITH CARE

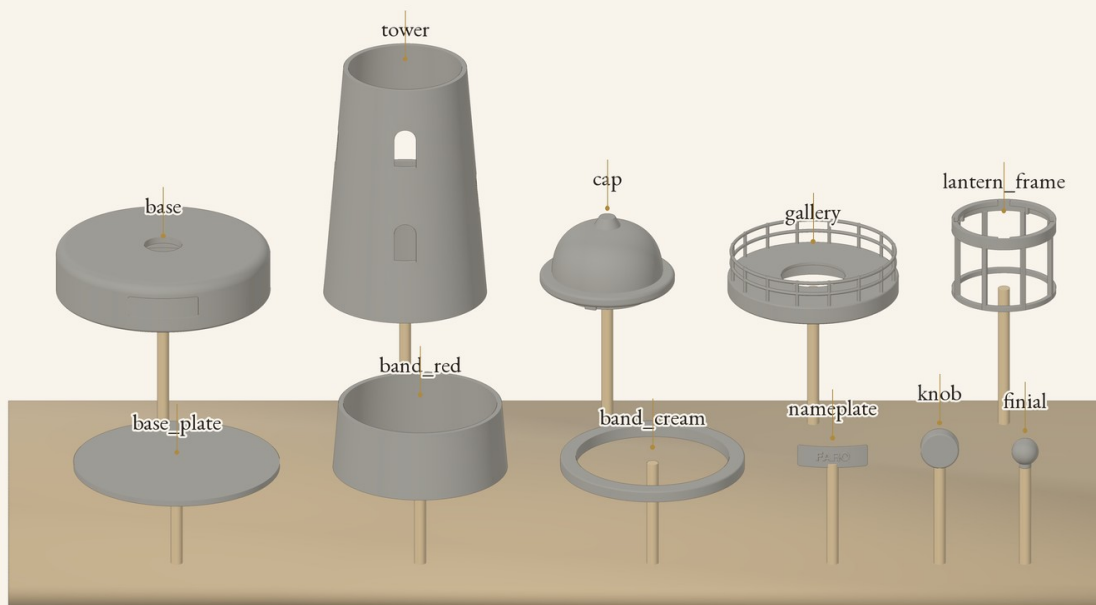
- Gallery and lantern frame: only remove support marks; the posts are delicate.
- Nameplate: sand the flat face only, never across the raised letters.
- Lantern glass and diffusers: never sand; they arrive frosted. Set them aside, clean.

STEP 03

Prime

Primer fills fine layer lines and shows every flaw. This is where the final finish is really made.

- 01 Mount parts on a tube, bottle or cocktail sticks with blu-tack, so you can reach all round without touching.
- 02 Choose a dry, still day, around 15 to 25°C. Shake the primer for two to three minutes.
- 03 Spray light coats from 25 to 30 cm, starting and finishing each pass off the part.
- 04 Let it dry, then sand with P400. Repeat until the surfaces look smooth under a bright light.
- 05 Prime every part except the lantern glass and diffusers. On the gallery and frame, one or two mist coats only.



Every part mounted for priming, except the glass and diffusers.

STEP 04

Paint

*Every part is a single colour, so no masking between colours is needed.
Tape only the surfaces that will be glued.*

COLOUR	PARTS
Red gloss	band_red, cap
Cream gloss	band_cream, tower
Metallic gold	gallery, lantern_frame, knob, finial, nameplate
Burnt umber	base, sponged wood effect
Black	base_plate, and the recesses round the letters



SPRAYING

- 01 Three or four light coats, 25 to 30 cm away, waiting between coats as the can directs.
- 02 On the tower, spray at an angle into each window so the edges are covered.
- 03 On the gallery and frame, use short bursts from several directions.
- 04 If dust or a run appears, let it dry, sand with P1200 and respray lightly.
- 05 For richer brass, spray gloss black first and gold over it.

COLOUR REFERENCE



STEP 04, CONTINUED

Wood and lettering

WALNUT BASE

- 01 Brush a thin coat of burnt umber over the primed base and let it dry.
- 02 Dab a barely loaded sponge on kitchen paper, then drag it round the base in one direction. The streaks read as grain.
- 03 Build two or three passes until it reads as dark walnut. Unevenness looks natural.

NAMEPLATE LETTERS

Once the gold is dry, brush thinned black acrylic over the letters, then wipe the raised surfaces with a damp cloth before it dries. The black stays round the letters and makes FARO easy to read.



Brush black over the letters.



Wipe the raised surface: FARO stands out in gold.

STEP 05

Clear coat and cure

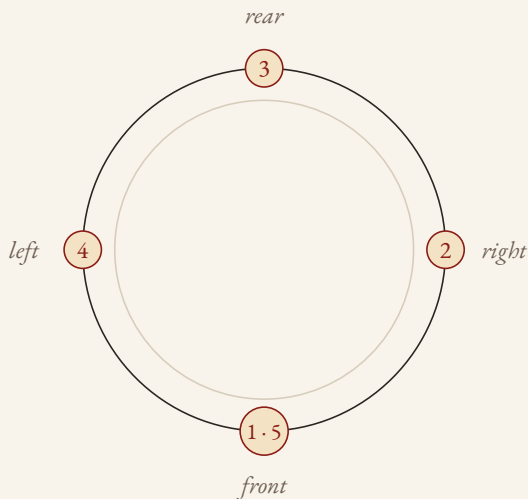
- 01 Wait at least 24 hours after the last colour coat.
- 02 Spray two light coats of clear lacquer on the red and cream parts only: band_red, cap, band_cream and tower.
- 03 Leave the gold and the wood-effect base uncoated; clear can dull metallic paint.
- 04 Leave everything warm, dry and dust-free for two to three days. Paint feels dry long before it is hard.
- 05 Optional: once hard, polish the red and cream parts with car polishing compound.

STEP 06

Assemble

Araldite Rapid sets in about five minutes. Mix small amounts, one joint at a time, and hold or tape each joint as it sets.

DIFFUSER MAP, SEEN FROM ABOVE



NO.	WINDOW	HEIGHT
1	Front, lowest	113 mm
2	Right side (seen from the front)	126 mm
3	Rear	139 mm
4	Left side	152 mm
5	Front, highest	165 mm



Each diffuser is numbered; hold it to its window first.

STEP 06, CONTINUED

Assembly order

- 01 Diffusers into the tower. Through the open top, glue each behind its window with tiny dabs at the edges only.
- 02 Nameplate into its recess on the front of the base.
- 03 Knob onto the front of the red band, centred.
- 04 Thread the lights in through the USB-C port at the back of the base and up through the hole in its top. The battery box stays outside, behind the lamp.
- 05 Stack and glue the cream band onto the base, then the red band, then the tower, with knob and nameplate aligned. Pull the lights up as you go.
- 06 Coil the lights loosely inside the tower so some sit near each window. Keep them all in the tower.
- 07 Gallery onto the tower top.
- 08 Lantern. Stand the empty glass on the gallery, lower the frame over it and glue the frame to the gallery. The lantern stays empty for the brightness test.
- 09 Finial into the cap, then twist the cap on, clockwise to the stop. Do not glue it.
- 10 Weight. Tape a few coins low inside the base.
- 11 Underneath. Glue the base plate into its rebate, then apply the felt, trimmed to size.

UNDERNEATH: FELT ON, AND LIFTED



The finished underside. In the prototype the plate is glued and the felt simply stuck on.

STEP 06, IN PICTURES

Assembly, step by step



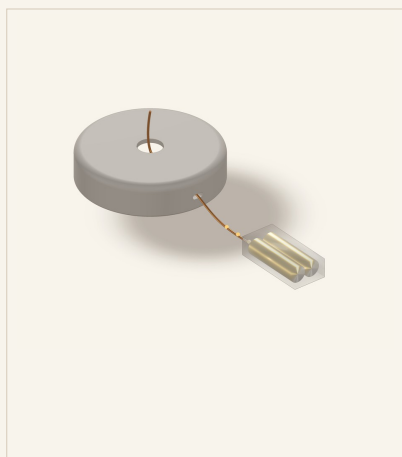
1 Diffusers in



2 Nameplate



3 Knob



4 Thread the lights



5 Cream band



5 Red band



5 Tower



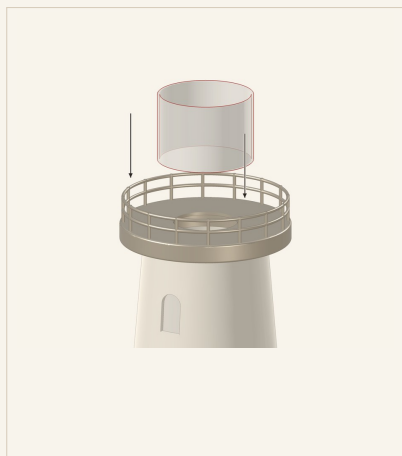
6 Coil the lights



7 Gallery

STEP 06, IN PICTURES, CONTINUED

Assembly, step by step



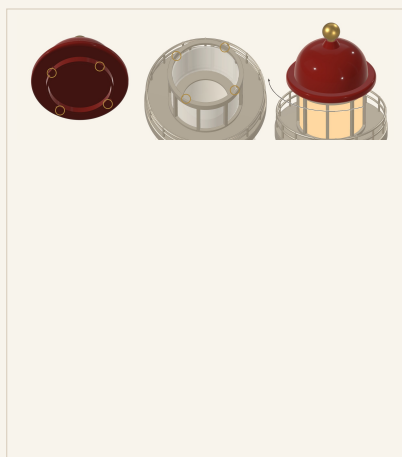
8 Lantern glass



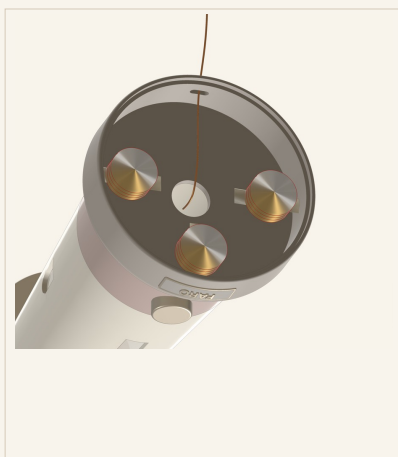
8 Lantern frame



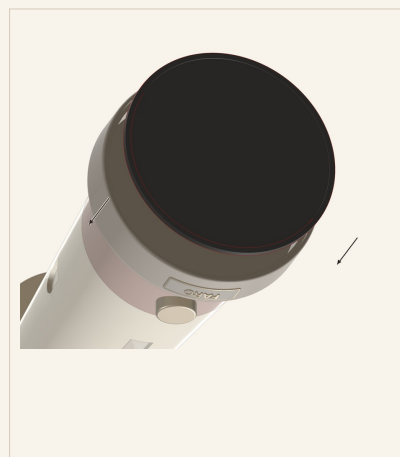
9 Finial



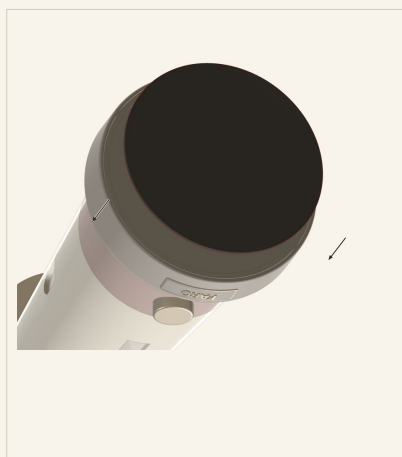
9 Cap on, clockwise



10 Weight



11 Base plate



11 Felt



Inside, finished



Finished

STEP 07

Light it and photograph

The fairy lights make the windows glow, with about the light of a couple of candles. The real lamp's light would come from an LED in the lantern, so test that too.

TESTING REAL BRIGHTNESS

- 01 Twist off the cap. The opening is 49 mm across.
- 02 Drop a warm white LED puck, under 45 mm across, into the empty lantern glass. Or run a USB LED module's cable in through the USB-C port and up to a power bank.
- 03 Twist the cap back on and switch off the room lights.
- 04 Judge it, and note what you see for the lighting ODM.

WHAT GOOD LOOKS LIKE

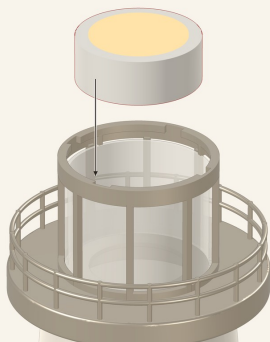
- It softly lights the nearby wall and surface, like a mood lamp.
- The frosted glass is bright without glaring.
- The light feels warm and domestic, not like a task light.

PHOTOGRAPHS

- Two sets: soft daylight with the lamp off; dusk with the lamp on.
- A plain backdrop, plus one styled shelf scene with books and a plant.
- Front, side, three-quarter, rear, lantern close-up, nameplate.
- Hide the battery box, power bank and cables out of shot.



Inside: the lights coiled in the tower, the puck in the empty lantern, coins in the base.



The puck rests on the ledge inside the gallery.



The goal: a soft, warm glow.

REFERENCE

Troubleshooting

Almost every problem is recoverable. Let paint harden fully before sanding or recoating.

PROBLEM	FIX
Cap won't twist on, or sticks	Sand the lugs and the underside of the slots lightly with P400, and test before adding more paint.
Paint run or sag	Let it harden, sand flat with P1200 and spray a light coat over it.
Rough, bumpy gloss (orange peel)	Sand gently with P1200 and polish, or add another light clear coat. Next time, lighter coats.
Dust specks	Once hard, sand the speck with P1200 and respray lightly on a still day.
Paint clogged in the railing	Clear the gaps with a cocktail stick while still wet. Once dry, don't force the posts.
Bright LED dots in a window	Stick a small piece of baking paper behind that diffuser, or move the lights slightly away.
A window stays dark	Move a coil of lights closer to it through the cap opening.
Lantern too dim, or glaring	Try a brighter or dimmer puck, or move it lower in the glass. Note the change for the ODM.
Lamp wobbles	Check the felt is flat and fully stuck, and add more coins low in the base.

REFERENCE

Test record

Write down what you learn before changing anything. It becomes the brief for the production lighting partner.

LED used			
Power source			
Distance to wall			
Brightness	☐ Too dim	☐ About right	☐ Too bright
Warmth	☐ Too cool	☐ About right	☐ Too warm
Glare	☐ None	☐ Mild	☐ Distracting
Diffusion	☐ Even	☐ Some hot spots	☐ Hot spots
Wall glow	☐ Weak	☐ Soft mood light	☐ Strong

NOTES FOR THE LIGHTING ODM

BEFORE YOU FINISH

- ☐ Cap still twists off
- ☐ All five windows glow
- ☐ Lantern glass empty when not testing
- ☐ Battery box and cables hidden in photos
- ☐ Changes noted for prototype two